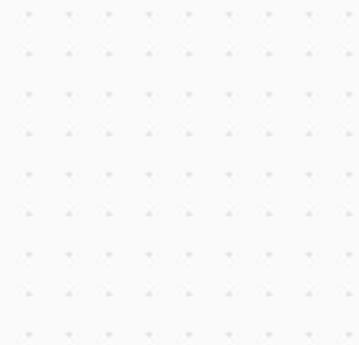




Causal Delay Copilot

The layer that tells you why a delay is coming – and what to do about it.

Track 2 – Supply Chain · Kaya AI India Hackathon 2026

98%

of megaprojects run late or over budget.

Large projects typically take **20% longer** than planned.

Up to **80% over budget**.

77% are at least 40% behind schedule.

Source: McKinsey Global Institute research.

"We are great at spotting risk. We are still terrible at understanding it."



A shipment is flagged "at risk." Now what?

A project manager sees a red alert on a dashboard.

The tool says delay likely.

It does not say why.

The PM scrambles through emails, call logs, and spreadsheets.

Guesswork replaces data – and the clock is ticking.



Predictive tools answer “what.” We answer “why.”

✓ What they do well

Learn patterns from past data

Flag anomalies early

Reduce surprise

What they can't do

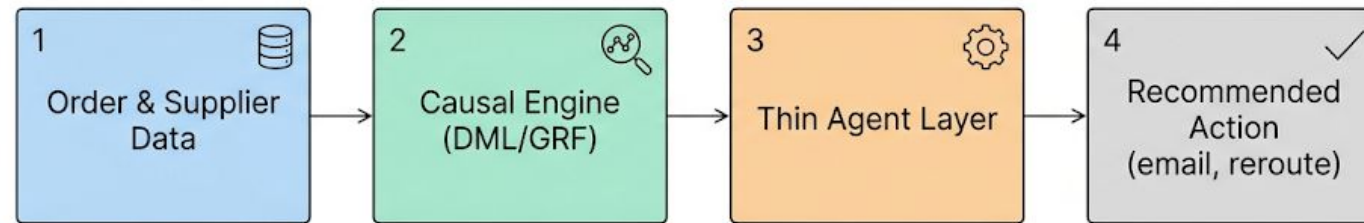
Separate true causes from coincidences

Tell you this delay is **because of X**

Hand you a concrete next action

Causal Delay Copilot fills exactly that gap.

How It Works – One Flow



1

Construction-flavoured
order & supplier data



2

Causal engine estimates
true
delay driver and effect size



3

A thin agent layer drafts one
recommended action tied to the
cause

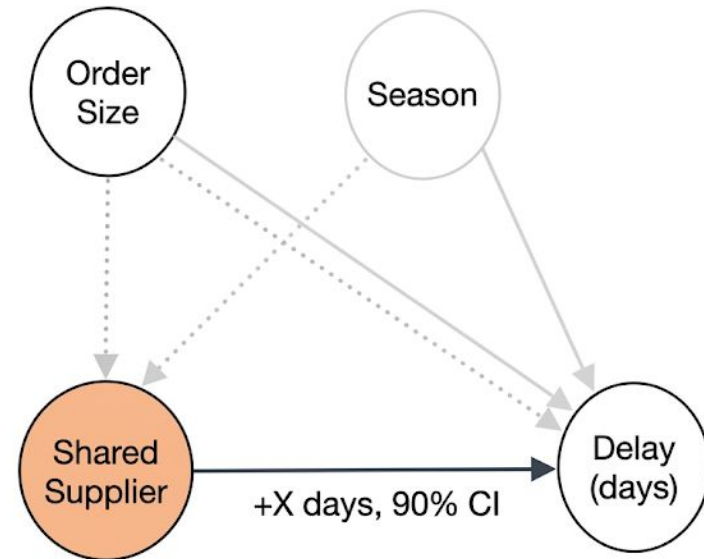
Not Another Predictive Model. A Causal One.

Built with Double Machine Learning / Generalized Random Forests (EconML).

"Does a shared supplier causally increase delay, after accounting for order size, material, and season?"

Unlike correlation, this isolates the **true effect** of a single factor.

 Research-backed: Wyrembek et al. (2024)



An Agent That Acts on the Explanation

1

Monitor

Node flags an at-risk order based on incoming data streams.

2

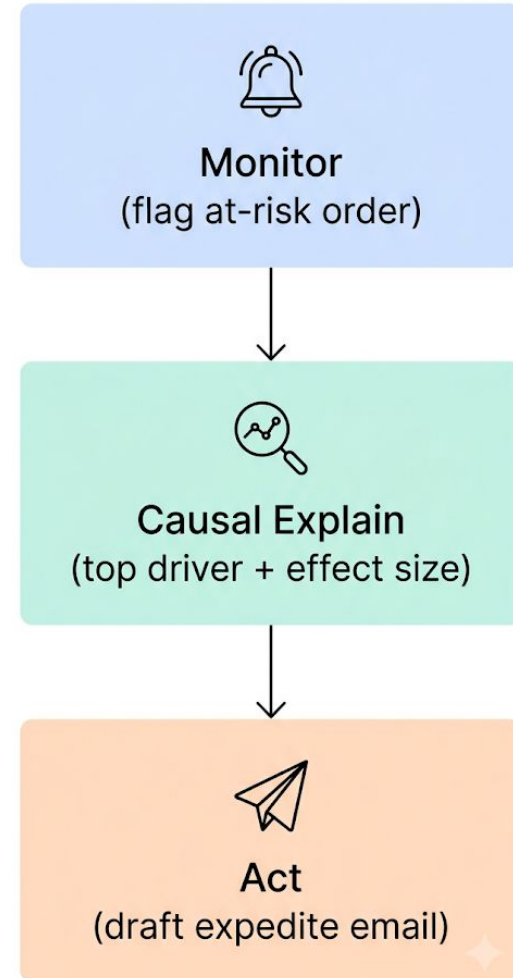
Explain

Causal Explain node returns top delay driver + effect in days.

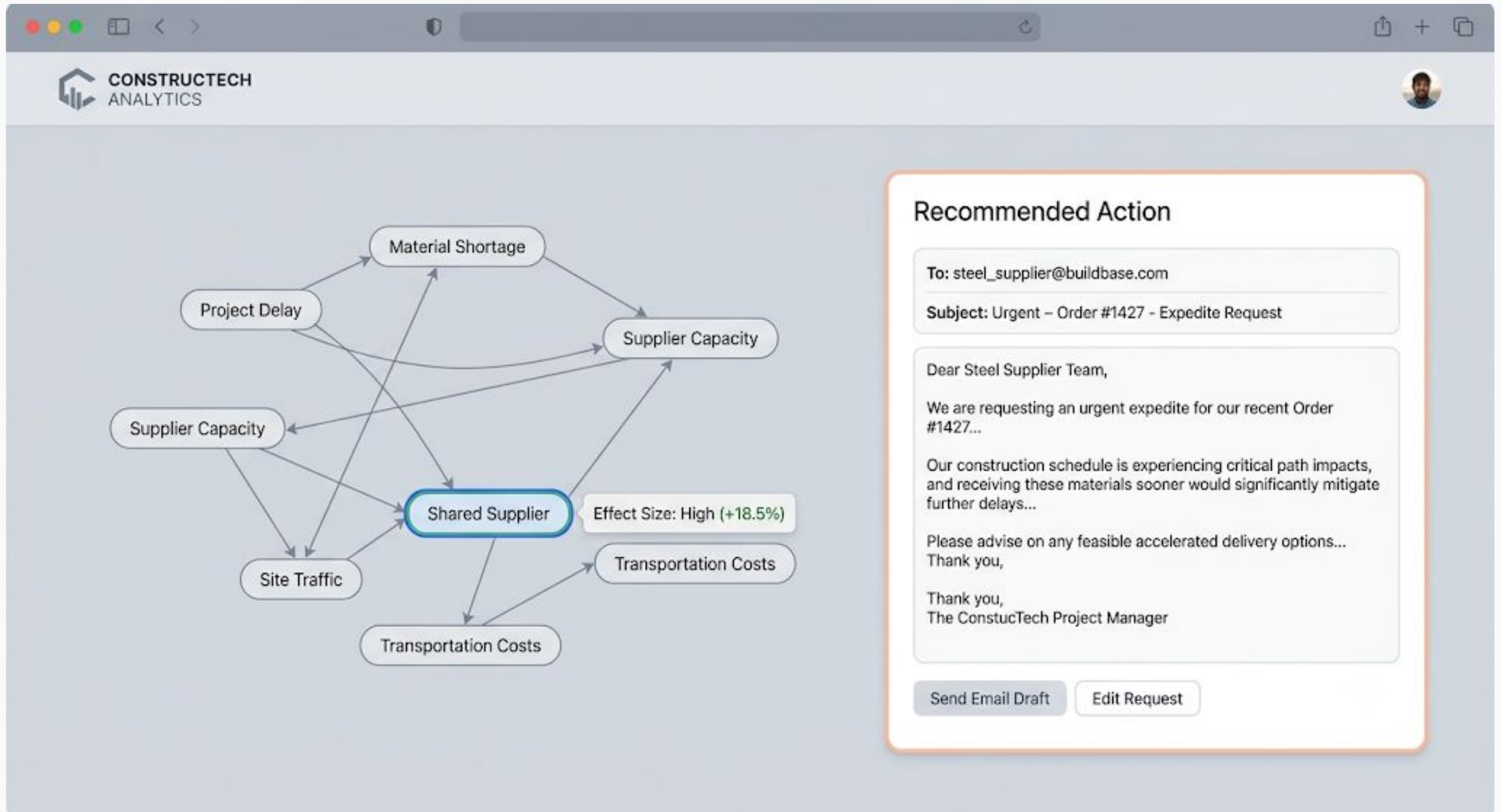
3

Act

Action node drafts one concrete artefact: expedite email, alternate vendor suggestion, or PM escalation note.



What the PM Sees Instead of a Red Flag



Built to Scale Across Projects and Delay Drivers



Generalizable Pipeline

The same DML pipeline works for any delay factor: weather exposure, single-sourcing, seasonal ordering patterns.



Data Adaptive

No re-architecting needed – just retrain on relevant data.



Lightweight Architecture

Designed as a lightweight layer that can sit on top of existing supply chain platforms, not a replacement.



Fits naturally into tools like Kaya's Amber and Jarvis, adding the explanatory **"why"** on top of their predictive **"what."**

Team & What's Next

Mridul Sharma

IIT Bombay



Conformal prediction for
calibrated uncertainty intervals.

Sneha Shandilya

IIT Bombay



Document ingestion pipeline
(drawings, specs, schedules).

Diggaj Maheshwari

IIT Bombay



Voice/WhatsApp layer for field
workers.

“Causal Delay Copilot – the layer that turns risk into a reason.”